Assignment 3

Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name: string, price: integer)

Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

For the above schema, perform the following—

1-Create the tables with the appropriate integrity constraints.

Query 1 SQL File 3*

Limit to 1000 rows

```sql
create database Assignment_3;
use Assignment_3;
#ques_1;
create table customer(
custo_id int primary key,
cust_name varchar(50) not null);

CREATE TABLE Item (
    item_id INT PRIMARY KEY,
    item_name VARCHAR(50) NOT NULL,
    price INT CHECK (price > 0)
);
 CREATE TABLE SALE(
 BILL_NO INT PRIMARY KEY,
 BILL_DATE DATE NOT NULL,
 custo_id INT,
 item_id int,
 qty_sold int,
foreign key (custo_id) references customer(custo_id),
foreign key (item_id) references item(item_id)
);
```

2-Insert around 10 records in each of the tables.

```
#ques_2
INSERT INTO customer values
(1,'Riya'),
(2,'Aman'),
(3,'Sita'),
(4,'Rahul'),
(5,'Neha'),
(6,'Karan'),
(7,'Meena'),
(8,'Arjun'),
(9,'Priya'),
(10,'Rohit');
INSERT INTO item values
(101,'Pen',50),
(102,'Notebook',120),
(103,'Bag',500),
(104,'Bottle',250),
(105,'Shoes',1500),
(106,'Watch',2200),
(107,'Mouse',300),
(108,'Keyboard',800),
(109,'Book',350),
(110,'Charger',600);
INSERT INTO SALE values
```

```
(106,'Watch',2200),
(107,'Mouse',300),
(108,'Keyboard',800),
(109,'Book',350),
(110,'Charger',600);
INSERT INTO SALE values
(1,CURDATE(),1,101,2),
(2,CURDATE(),2,103,1),
(3,CURDATE(),5,104,3),
(4,'2026-02-20',3,105,1),
(5,'2026-02-19',5,106,1),
(6,CURDATE(),6,107,4),
(7,CURDATE(),7,108,2),
(8,'2026-02-18',8,109,1),
(9,CURDATE(),9,110,2),
(10,CURDATE(),5,102,5);
```

```
(5,'2026-02-19',5,106,1),
(6,CURDATE(),6,107,4),
(7,CURDATE(),7,108,2),
(8,'2026-02-18',8,109,1),
(9,CURDATE(),9,110,2),
(10,CURDATE(),5,102,5);
select * from customer;
```

custo_id	cust_name
1	Riya
2	Aman
3	Sita
4	Rahul
5	Neha
6	Karan
7	Meena
8	Arjun
9	Priya
10	Rohit

```
(6,CURDATE(),6,107,4),
(7,CURDATE(),7,108,2),
(8,'2026-02-18',8,109,1),
(9,CURDATE(),9,110,2),
(10,CURDATE(),5,102,5);
select * from customer;
select * from item;
```

item_id	item_name	price
101	Pen	50
102	Notebook	120
103	Bag	500
104	Bottle	250
105	Shoes	1500
106	Watch	2200
107	Mouse	300
108	Keyboard	800
109	Book	350
110	Charger	600

```
(7,CURDATE(),7,108,2),
(8,'2026-02-18',8,109,1),
(9,CURDATE(),9,110,2),
(10,CURDATE(),5,102,5);
select * from customer;
select * from item;
select * from SALE;
```

BILL_NO	BILL_DATE	custo_id	item_id	qty_sold
1	2026-02-21	1	101	2
2	2026-02-21	2	103	1
3	2026-02-21	5	104	3
4	2026-02-20	3	105	1
5	2026-02-19	5	106	1
6	2026-02-21	6	107	4
7	2026-02-21	7	108	2
8	2026-02-18	8	109	1
9	2026-02-21	9	110	2
10	2026-02-21	5	102	5

3-List all the bills for the current date with the customer names and item numbers.

```
select * from customer;
select * from item;
select * from SALE;
#ques_3
SELECT S.BILL_NO, C.cust_name, S.item_id
FROM SALE S
JOIN customer C ON S.custo_id = C.custo_id
WHERE S.BILL_DATE = CURDATE();
```

BILL_NO	cust_name	item_id
1	Riya	101
2	Aman	103
3	Neha	104
6	Karan	107
7	Meena	108
9	Priya	110
10	Neha	102

4-List the total Bill details with the quantity sold, price of the item and the final amount.

```
#ques_4
SELECT S.BILL_NO,
       S.qty_sold,
       I.price,
       (S.qty_sold * I.price) AS Final_Amount
FROM Sale S
JOIN Item I ON S.item_id = I.item_id;
```

BILL_NO	qty_sold	price	Final_Amount
1	2	50	100
2	1	500	500
3	3	250	750
4	1	1500	1500
5	1	2200	2200

5- List the details of the customer who have bought a product which has a price>200.

```
        I.price,
        (S.qty_sold * I.price) AS Final_Amount
FROM Sale S
JOIN Item I ON S.item_id = I.item_id;
#ques_5;
select DISTINCT c.custo_id,c.cust_name
from  customer c
join sale s on c.custo_id=s.custo_id
join item i on s.item_id=i.item_id
where i.price>200;
```

custo_id	cust_name
2	Aman
5	Neha
3	Sita
6	Karan
7	Meena
8	Arjun
9	Priya

6-Give a count of how many products have been bought by each customer.

```
join item i on s.item_id=i.item_id
where i.price>200;
#ques_6;
SELECT c.custo_id,
       c.cust_name,
       COUNT(s.item_id) AS Total_Products_Bought
FROM customer c
JOIN sale s ON c.custo_id = s.custo_id
GROUP BY c.custo_id, c.cust_name;
```

custo_id	cust_name	Total_Products_Bought
1	Riya	1
2	Aman	1
3	Sita	1
5	Neha	3
6	Karan	1
7	Meena	1
8	Arjun	1
9	Priya	1

7- Give a list of products bought by a customer having cust_id as 5.

```
#ques_7;
SELECT i.item_name
FROM item i
JOIN sale s ON i.item_id = s.item_id
WHERE s.custo_id = 5;
```

item_name
Bottle
Watch
Notebook

8- List the item details which are sold as of today.

```
#ques_8;
SELECT DISTINCT i.*
FROM item i
JOIN sale s ON i.item_id = s.item_id
WHERE s.BILL_DATE = CURDATE();
```

item_id	item_name	price
101	Pen	50
103	Bag	500
104	Bottle	250
107	Mouse	300
108	Keyboard	800
110	Charger	600
102	Notebook	120

9- Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold, amount.

```
#ques_9;
CREATE VIEW Bill_View AS
SELECT s.BILL_NO,
       s.BILL_DATE,
       s.custo_id,
       s.item_id,
       i.price,
       s.qty_sold,
       (i.price * s.qty_sold) AS amount
FROM sale s
JOIN item i ON s.item_id = i.item_id;
select * from Bill_View;
```

Result Grid

BILL_NO	BILL_DATE	custo_id	item_id	price	qty_sold	amount
1	2026-02-21	1	101	50	2	100
2	2026-02-21	2	103	500	1	500
3	2026-02-21	5	104	250	3	750
4	2026-02-20	3	105	1500	1	1500
5	2026-02-19	5	106	2200	1	2200
6	2026-02-21	6	107	300	4	1200
7	2026-02-21	7	108	800	2	1600
8	2026-02-18	8	109	350	1	350
9	2026-02-21	9	110	600	2	1200
10	2026-02-21	5	102	120	5	600

10- Create a view which lists the daily sales date wise for the last one week.

```
#ques-10;CREATE VIEW Weekly_Sales AS
SELECT s.BILL_DATE,
       SUM(i.price * s.qty_sold) AS Total_Daily_Sales
FROM sale s
JOIN item i ON s.item_id = i.item_id
WHERE s.BILL_DATE >= CURDATE() - INTERVAL 7 DAY
GROUP BY s.BILL_DATE;
```

Result Grid

BILL_DATE	Total_Daily_Sales
2026-02-21	5950
2026-02-20	1500
2026-02-19	2200
2026-02-18	350